

Results

Generated catalog, metrics, and outputs I

The live registry reports 17 implemented generators, 18 catalog entries, 18 evidence records, and 36 source records in the checked-in scholarship audit snapshot. The complete catalog is generated in Appendix A rather than copied into prose. The companion figure keeps the categorical registry, source namespace, and implementation statuses visible at a glance.

Generated catalog, metrics, and outputs I

Catalog status and evidence matrix

Taxonomy facets are orthogonal; status and evidence are separate dimensions.

ID	modality	mechanism	signature	input	status
visual.duck_rabbit	visual	ambiguity	bistability	none	implemented
visual.simultaneous_contrast	visual	contrast_and_context	contrast_distortion	none	implemented
visual.apparent_motion	visual	temporal_motion	illusory_motion	none	implemented
audio.shepard_tone	auditory	spectral_harmonic	continuity	none	implemented
audio.missing_fundamental	auditory	spectral_harmonic	filling_in	none	implemented
audiovisual.sound_induced_flash	audio_visual	crossmodal_temporal	fusion_or_fission	controlled_playback	implemented
audiovisual.ventriloquist	audio_visual	crossmodal_spatial	spatial_capture	stereo_playback	implemented
visual.muller_lyer	visual	geometric_alignment	geometric_distortion	none	implemented
visual.poggendorff	visual	geometric_alignment	geometric_distortion	none	implemented
visual.ponzo	visual	geometric_alignment	geometric_distortion	none	implemented
visual.kanizsa_triangle	visual	contrast_and_context...	filling_in	none	implemented
visual.ebbinghaus	visual	contrast_and_context...	geometric_distortion	none	implemented
visual.zollner	visual	geometric_alignment	geometric_distortion	none	implemented
audio.tritone_paradox	auditory	spectral_harmonic	categorical_recoding	none	implemented
audio.octave_illusion	auditory	stream_segregation	fusion_or_fission	stereo_playback	implemented
audio.auditory_continuity	auditory	stream_segregation	continuity	none	implemented
audiovisual.temporal_ventriloquism	audio_visual	crossmodal_temporal	temporal_binding_or_recalibration	controlled_playback	implemented
audiovisual.mcquirk	audio_visual	speech_categorization	categorical_recoding	validated_speech_feature	input_required

Generated catalog, metrics, and outputs I

What this figure shows: A matrix lists all catalog entries with modality, mechanism, signature, input requirement, and implementation status. The live catalog matrix contains 18 entries and keeps modality, mechanism, perceptual signature, input requirement, evidence status, and implementation status as separate facets. Source data output/data/catalog_matrix.json preserve the exact taxonomy snapshot and its source-reference namespace; status is not a proxy for perceptual validation. Controls: live taxonomy registry; evidence matrix snapshot; seed not applicable to the catalog diagram. Objective facts: 18 catalog rows; categorical facets and statuses; no observer-level quantity is applicable. Claim level: source_supported. Source data: output/data/catalog_matrix.json (SHA-256 digest recorded in the figure registry). Evidence lineage: gregory1997visual, hirst2020sound, bruns2019ventriloquist. Limitations: Taxonomy facets are engineering

classifications and remain provisional where the literature supports competing accounts. Boundary: Catalog coverage is bounded to this registered package and does not enumerate all known illusions.

The complete catalog matrix fig. 1 is tabulated in the standalone Appendix A sec. 10 and in tbl. 10. The corresponding machine-readable evidence matrix records the source role, exact supported claim, engineering departure, and limitation for every entry. Keeping the table in an appendix gives the main results narrative room to explain the contract without turning a mutable registry snapshot into a prose claim.

Typed domains and delivery profiles I

Typed domains and delivery profiles I

Table 1: Typed parameter domains and units.

Type	Unit	Validated domain	Role
PixelDimension	pixels	8–4096	image/video width and height
GrayscaleLevels	levels	2–256	representable grayscale levels
QuantizationLevels	levels	2–256	output quantization buckets
FrequencyHz	Hz	> 0 and finite	oscillator and harmonic frequencies
SampleRate	samples/s	1,000–384,000	audio sampling clock
FrameRate	frames/s	1–240	video presentation clock
SyncOffsetMs	ms	–10,000–10,000	audio relative to video
SpatialOffset	normalized	–1–1	declared crossmodal spatial discrepancy

Typed domains and delivery profiles I

The parameter contracts are summarized in tbl. 1. They constrain physical construction and serialization; they are not psychophysical scales.

Typed domains and delivery profiles I

Table 2: Encoding profiles and backend capabilities.

Format	Backend	Artifact	Profile	Status
PNG	Pillow	image	lossless uint8 raster	available
WAV	python wave	audio	PCM 8/16/24/32-bit	available
GIF	Pillow	video	palette animation with declared duration	available
NPZ	NumPy	canonical artifact	exact little-endian float32 archive	available
MP4	ffmpeg	video/audiovisual	H.264 or muxed MP4; decode inspected	available

Typed domains and delivery profiles I

The delivery profiles are summarized in tbl. 2. Backend availability is environment-dependent and is recorded at generation time.

The catalog distinguishes a physical construction from an observer claim. For example, the sound-induced-flash generator establishes one flash and one or two beep events on a shared clock; it does not establish a reported flash count. Similarly, the temporal-ventriloquism generator exposes a declared timing offset, while the literature reports observer-dependent temporal judgments under specific tasks and timing conditions (Vroomen and Gelder 2004).

Pipeline and provenance I

Pipeline and provenance I

DuckRabbit reproducibility contract

Canonical identity is established before delivery; decoded inspection tests what was actually written.



Engineering trace only: deterministic media facts and verification are not observer evidence.

Figure 2: Five labeled stages connect a typed request to a deterministic canonical artifact, encoded media, decoded inspection, and a verification manifest.

Pipeline and provenance I

What this figure shows: Five labeled stages connect a typed request to a deterministic canonical artifact, encoded media, decoded inspection, and a verification manifest. This schematic separates a typed request from the canonical artifact it generates, the optional delivery encoder, decoded inspection, and the manifest-level verification decision. Arrows indicate data and provenance dependencies rather than a causal model of perception; the seed is 0 and the complete stage record is in source data output/data/architecture.json. Controls: typed request $q=(i, s, e)$ with seed $s=0$; default generator and delivery-independent canonicalization. Objective facts: no unit-bearing objective quantity is applicable to this architecture diagram; stage identities and provenance fields are recorded in the sidecar. Claim level: canonical_stimulus. Source data: output/data/architecture.json (SHA-256 digest recorded in

the figure registry). Evidence lineage: `formalism:eq:typed_request`, `formalism:eq:canonical_generation`, `formalism:eq:encoding_verification`. Limitations: The schematic abstracts implementation boundaries and does not represent an observer study or a causal theory of perception. Boundary: The pipeline verifies reproducible media facts, not what a person experiences.

The generated architecture figure fig. 2 shows the separation between request, canonical artifact, delivery adapter, decoded inspection, and manifest. The canonical digest is the identity of the in-memory stimulus; an encoded hash is the identity of the delivered file. The difference is intentional.

Visual constructions and parameter sweeps I

Visual constructions and parameter sweeps I

Complete panel of implemented visual constructions

One typed default per implemented entry. Static families show canonical rasters; apparent motion shows frame 1, with sequence facts retained in source data.
Reading guide: luminance and raster levels describe pixels; frame rate and frame delta describe temporal media; neither is an observer report.

visual.duck_rabbit



canonical raster

Luminance mean: 0.777 · SD: 0.371

Levels: 3 · digest: dde9d5ecda...

visual.simultaneous_contrast



canonical raster

Luminance mean: 0.500 · SD: 0.328

Levels: 3 · digest: 696ba38bda...

visual.apparent_motion

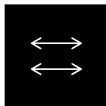


representative first frame; sequence in source data

Frames: 6 · rate: 12.0 frames/s

Frame delta: 0.242 normalized · digest: acc31937b4...

visual.muller_lyer



canonical raster

Luminance mean: 0.027 · SD: 0.162

Levels: 2 · digest: 3e97a33e27...

visual.poggendorff



canonical raster

Luminance mean: 0.989 · SD: 0.106

Levels: 2 · digest: 21d6b959d4...

visual.ponzo



canonical raster

Luminance mean: 0.958 · SD: 0.201

Levels: 2 · digest: e5a04c8921...

visual.kanizsa_triangle



canonical raster

Luminance mean: 0.921 · SD: 0.270

Levels: 2 · digest: db5f2b297b...

visual.ebbinghaus



canonical raster

Luminance mean: 0.347 · SD: 0.152

Levels: 3 · digest: 1560d41307...

visual.zollner



canonical raster

Luminance mean: 0.937 · SD: 0.242

Levels: 2 · digest: 917daed5dc...

Visual constructions and parameter sweeps I

What this figure shows: A 9-panel gallery shows every currently implemented visual catalog entry, with stable IDs and physical raster summaries: Duck-rabbit ambiguous figure, Simultaneous contrast stimulus, Apparent-motion frame sequence, Müller-Lyer geometric illusion, Poggendorff geometric illusion, Ponzo perspective illusion, Kanizsa illusory-contour triangle, Ebbinghaus context-size illusion, Zöllner orientation illusion. This complete gallery presents one deterministic representative from each of the 9 currently implemented visual catalog entries: Duck-rabbit ambiguous figure, Simultaneous contrast stimulus, Apparent-motion frame sequence, Müller-Lyer geometric illusion, Poggendorff geometric illusion, Ponzo perspective illusion, Kanizsa illusory-contour triangle, Ebbinghaus context-size illusion, Zöllner orientation illusion. The panel therefore covers the package's available visual families—ambiguous figure, contrast, apparent motion, geometric

Visual constructions and parameter sweeps I

context, illusory contour, contextual size, and orientation-related constructions—using typed defaults at seed 0. For the temporally expressed apparent-motion entry, the displayed tile is its first frame and the source data preserve the full sequence. The source data sidecar `output/data/visual_panel.json` records the stable ID, generator parameters, canonical SHA-256 digest, Rec. 709 or grayscale luminance statistics, raster-level counts, and parameter boundary for every tile. Controls: one default typed parameter object per implemented visual entry; seed `s=0`; dimensions, luminance bounds, grayscale, and quantization controls remain in each generator schema; temporal entries display their first frame but retain sequence metrics. Objective facts: per tile: mean and standard-deviation relative luminance, unique raster levels, and canonical SHA-256 digest; temporal entries additionally retain frame count, frame rate, and frame deltas; definitions and units

Visual constructions and parameter sweeps I

are in the source data JSON. Claim level: canonical_stimulus. Source data: output/data/visual_panel.json (SHA-256 digest recorded in the figure registry). Evidence lineage: gregory1997visual, brugger1999duckrabbit, howe2005muller, morgan1999poggendorff, fisher1967ponzo, yildiz2022ponzo, kanizsa1976contours, mruczek2015ebbinghaus, earle1995zollner, wertheimer1912motion, sekuler1996wertheimer. Limitations: The gallery is complete for implemented visual entries in this package, not a complete survey of visual illusions; literature families, historical displays, and DuckRabbit rasters are not pixel-identical by default. Viewing scale, display calibration, viewing distance, and observer conditions can change the relevance of a construction; the gallery does not measure an observer's perceptual report. Boundary: The figure establishes coverage of the package's implemented visual constructions and their

deterministic raster facts. It does not establish that any viewer will perceive the named signature, nor that the set is exhaustive of the visual-illusion literature.

The visual panel fig. 3 is deliberately a coverage figure: it contains one deterministic representative for every currently implemented visual entry, including apparent motion and Zöllner in addition to the static ambiguous-figure, contrast, geometric-context, illusory-contour, and contextual- size families. Equalities, masks, line geometry, luminance bounds, and temporal positions are properties of the typed rasters or sequences. They are not observer scores, and the gallery is not an exhaustive ontology of visual illusions.

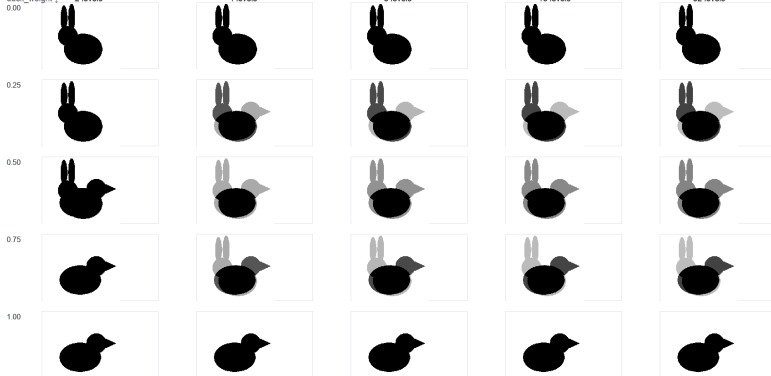
Visual constructions and parameter sweeps I

Duck-rabbit parameter sweep

Rows vary blend weight; columns vary grayscale and quantization levels. Each cell is a physical construction.

grayscale / quantization →

duck_weight ↓ 2 levels



The sweep reports canonical digests, unique levels, luminance, and dynamic range; it does not estimate perceptual sensitivity.

Figure 4: A labeled grid crosses five duck-rabbit blend weights with five grayscale and quantization settings, with physical metrics recorded for each cell.

Visual constructions and parameter sweeps I

What this figure shows: A labeled grid crosses five duck-rabbit blend weights with five grayscale and quantization settings, with physical metrics recorded for each cell. This 5×5 sweep varies duck-rabbit blend weight across rows and jointly varies grayscale and quantization levels across columns. Each cell is regenerated from an immutable typed parameter object at seed 0; source data output/data/visual_sweep.json records canonical digests, unique-level counts, normalized luminance, and effective dynamic range for every condition. Controls: duck_weight {0,.25,.5,.75,1}; grayscale=quantization {2,4,8,16,32}; seed s=0. Objective facts: 25 canonical rasters; unique levels, normalized mean luminance, and effective dynamic range per cell. Claim level: physical_metric. Source data: output/data/visual_sweep.json (SHA-256 digest recorded in the figure registry). Evidence lineage: brugger1999duckrabbit, gregory1997visual. Limitations: The grid is a

Visual constructions and parameter sweeps I

sensitivity of the construction parameters, not a psychophysical sensitivity curve or a validated observer effect. Boundary: The chosen grid is an engineering sampling of the parameter domain and does not imply an optimal or perceptually uniform spacing.

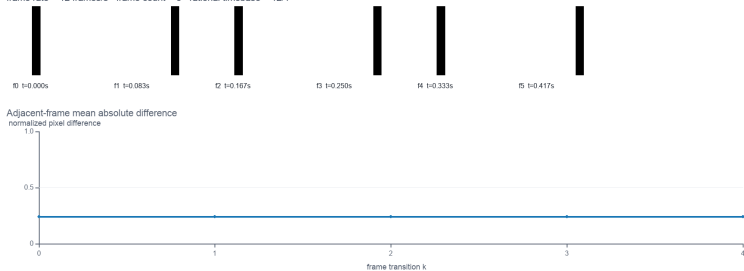
The sweep fig. 4 varies duck-weight from 0 to 1 and jointly varies representable grayscale and quantization levels. The source data preserve canonical digests and unique-value counts; no perceptual score is assigned to a sweep cell.

Temporal and auditory constructions I

Temporal and auditory constructions I

Temporal sequence and frame deltas

frame rate = 12 frames/s · frame count = 6 · rational timebase = 12/1



Frame timestamps and deltas are decoded/canonical timing facts; apparent motion remains an observer-level hypothesis.

Figure 5: Six ordered frames are labeled with timestamps and paired with a line plot of adjacent-frame normalized pixel differences.

Temporal and auditory constructions I

What this figure shows: Six ordered frames are labeled with timestamps and paired with a line plot of adjacent-frame normalized pixel differences. The frame strip exposes the order and timestamps of the apparent-motion construction, while the lower trace reports adjacent-frame mean absolute pixel differences. Source data `output/data/temporal_sequence.json` preserve frame count, frame rate, reduced rational timebase, timestamps in seconds, and the nonzero transition series; seed 0 identifies the generated sequence. Controls: `visual.apparent_motion` default typed parameters; frame rate and frame count from the canonical sequence; seed `s=0`. Objective facts: timestamps in seconds, frame rate in frames/s, frame count, rational timebase, and adjacent-frame normalized pixel difference. Claim level: `physical_metric`. Source data: `output/data/temporal_sequence.json` (SHA-256 digest recorded in the figure registry). Evidence lineage:

Temporal and auditory constructions I

wertheimer1912motion, sekuler1996wertheimer. Limitations: Frame succession and frame deltas are physical file properties; playback timing, display persistence, and motion reports require a controlled observer task. Boundary: The figure documents temporal structure rather than the presence, direction, or strength of perceived motion.

The temporal figure fig. 5 reports frame succession, rational frame rate, and frame-to-frame mean absolute differences. These facts establish temporal structure in the file, not the presence or strength of perceived motion.

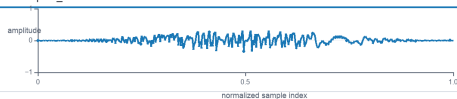
Temporal and auditory constructions I

Temporal and auditory constructions II

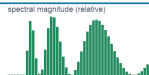
Auditory canonical signals

Canonical waveforms, one-sided spectral summaries, and typed playback facts; no pitch report is inferred.

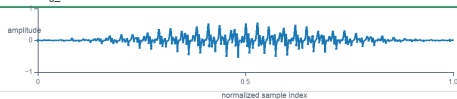
audio.shepard_tone



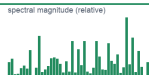
RMS 0.1128 normalized amplitude
peak 0.3480 normalized amplitude
centroid 1699.5 Hz
bandwidth 870.3 Hz · crest 3.09



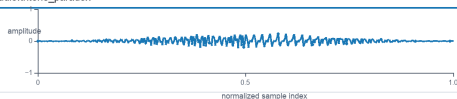
audio.missing_fundamental



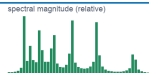
RMS 0.1474 normalized amplitude
peak 0.5573 normalized amplitude
centroid 685.7 Hz
bandwidth 240.4 Hz · crest 3.78



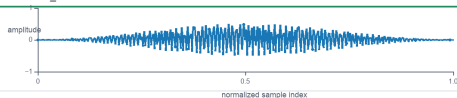
audio.tritone_paradox



RMS 0.0821 normalized amplitude
peak 0.2387 normalized amplitude
centroid 1019.2 Hz
bandwidth 715.3 Hz · crest 2.91



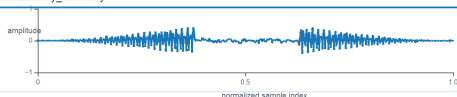
audio.octave_illusion



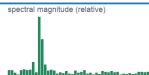
RMS 0.2164 normalized amplitude
peak 0.4999 normalized amplitude
centroid 510.8 Hz
bandwidth 623.4 Hz · crest 2.31



audio.auditory_continuity



RMS 0.1381 normalized amplitude
peak 0.4201 normalized amplitude
centroid 1333.1 Hz
bandwidth 1179.4 Hz · crest 3.04



Temporal and auditory constructions I

What this figure shows: Five labeled rows show canonical audio waveforms, relative one-sided spectral bars, RMS and peak in normalized amplitude, and spectral summaries in hertz. Five implemented auditory constructions—Shepard tone, missing fundamental, tritone paradox, octave illusion, and auditory continuity—are shown as canonical waveforms with compact one-sided FFT summaries. Source data `output/data/audio_signals.json` records sample rate, channel layout, RMS, peak, spectral centroid, bandwidth, crest factor, canonical digest, and the declared display-bin convention; no pitch or stream report is inferred. Controls: default typed audio parameters; per-generator channel layout and sample rate; seed `s=0`; one-sided FFT display with 48 sampled bins. Objective facts: RMS and peak in normalized amplitude; spectral centroid and bandwidth in Hz; sample rate in samples/s; channel layout and canonical SHA-256 digest. Claim level: `physical_metric`.

Temporal and auditory constructions I

Source data: `output/data/audio_signals.json` (SHA-256 digest recorded in the figure registry). Evidence lineage: shepard1984scale, zatorre2005missing, deutsch1986tritone, repp1997tritone, deutsch1974octave, warren1970continuity, rieke2011continuity. Limitations: Playback level, transducer response, dichotic separation, masker design, and listener judgments are external to the canonical buffer. Boundary: The figure compares generated signal properties and literature-linked families; it does not establish pitch, continuity, or stream organization for a listener.

Temporal and auditory constructions I

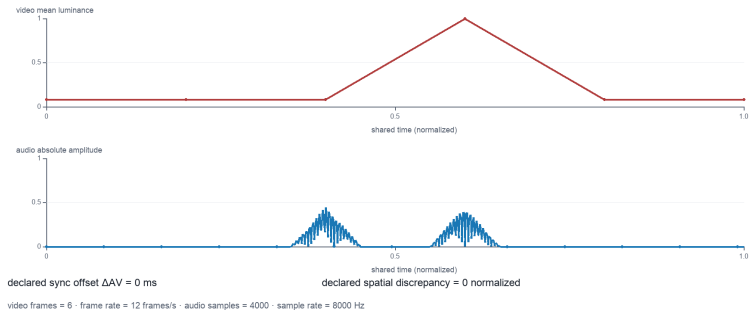
The auditory figure fig. 6 shows canonical waveforms and spectral summaries for Shepard, missing-fundamental, tritone, octave, and continuity constructions. RMS, peak, spectral centroid, bandwidth, and channel layout are calculated from the canonical buffer with explicit units and windowing rules. Pitch direction, pitch height, continuity, and stream organization remain observer-level outcomes that depend on playback and task conditions.

Audiovisual timing and encoding I

Audiovisual timing and encoding I

Audiovisual shared-clock timeline

Video and audio are plotted against normalized shared time; offsets are declared, not inferred from perception.



The aligned traces verify physical timing metadata; audiovisual capture or temporal recalibration requires a controlled observer task.

Figure 7: Two labeled traces share a normalized time axis: video mean luminance and audio absolute amplitude, with signed synchronization and spatial offsets printed below.

Audiovisual timing and encoding I

What this figure shows: Two labeled traces share a normalized time axis: video mean luminance and audio absolute amplitude, with signed synchronization and spatial offsets printed below. Video mean luminance and rectified audio amplitude are placed on a common normalized time axis for the sound-induced-flash construction. Source data output/data/audiovisual_timeline.json records the video frame clock, audio sampling clock, event traces, signed audio-minus-video synchronization offset in milliseconds, and normalized spatial discrepancy; alignment is declared by the generator, not estimated from perception. Controls: audiovisual.sound_induced_flash default shared timeline; declared sync offset ΔAV and spatial offset; seed $s=0$. Objective facts: video luminance and audio envelope on normalized shared time; ΔAV in ms; spatial discrepancy in normalized units; frame and sample clocks in the sidecar. Claim level: physical_metric. Source

Audiovisual timing and encoding I

data: output/data/audiovisual_timeline.json (SHA-256 digest recorded in the figure registry). Evidence lineage: shams2000sifi, hirst2020sound, vroomen2004temporal, hartcherobrien2011temporal. Limitations: Display refresh, audio hardware, event latency, and observer temporal binding are not measured by this figure. Boundary: A shared-clock construction is not evidence that observers bind the events at the declared or any other perceptual time.

The audiovisual timeline fig. 7 displays video luminance and audio amplitude against the shared clock, alongside declared synchronization and spatial offsets. Ventriloquist stimuli require stereo playback and spatially resolved display; temporal binding requires controlled timing and an observer task (Bruns 2019; Vroomen and Gelder 2004).

Audiovisual timing and encoding I

Encoding and verification surface

format	backend	canonical fact preserved	verification
PNG	Pillow	image shape + float32 digest	decoded dimensions
WAV	wave	sample count + clock	rate/channels/samples
GIF	Pillow	frame count + timing	decoded frames
NPZ	NumPy	exact arrays + digest	exact canonical round trip
MP4	ffmpeg	canonical artifact + stream facts	optional capability

Figure 8: A five-row comparison lists format, backend, preserved canonical facts, and the corresponding decoded verification checks.

Audiovisual timing and encoding I

What this figure shows: A five-row comparison lists format, backend, preserved canonical facts, and the corresponding decoded verification checks. The comparison separates the canonical artifact from delivery containers: PNG, WAV, GIF, NPZ, and optional MP4/muxed MP4. Source data output/data/encoding_verification.json records backend identity, preserved canonical facts, decoded inspection targets, and capability status as observed in the generation environment; the table therefore describes an adapter contract rather than a claim about codec quality. Controls: typed encoding profile selected per artifact; overwrite and backend capability checks; seed inherited from the stimulus request. Objective facts: format availability and verification scope are environment observations; NPZ is exact while other formats are decoded and tolerance-checked. Claim level: encoded_media. Source data: output/data/encoding_verification.json (SHA-256 digest recorded in the

figure registry). Evidence lineage:
engineering:typed_media_adapters. Limitations: Backend availability, codec versions, container metadata, and lossy tolerances are environment-dependent; playback software can expose additional behavior. Boundary: A successful encode or decode does not validate an observer effect or guarantee cross-device perceptual equivalence.

The encoding matrix fig. 8 makes backend capability and verification scope visible. NPZ is the exact canonical archive; PNG, WAV, GIF, and MP4 are delivery containers whose decoded facts are checked against the manifest.

Objective metric results I

Objective metric results I

Table 3: Objective metric definitions and epistemic boundary.

Metric	Unit	Definition	Tolerance	Claim level
mean_luminance	normalized_luminance	mean Rec. 709 relative luminance (grayscale is identity)	exact canonical	physical_metric
unique_values	levels	count of distinct canonical image values	exact integer	physical_metric
rms	normalized_amplitude	root-mean-square audio amplitude	finite scalar	physical_metric
peak	normalized_amplitude	maximum absolute audio amplitude	finite scalar	physical_metric
spectral_centroid_hz	Hz	energy-weighted one-sided spectral centroid	finite scalar	physical_metric
spectral_bandwidth_hz	Hz	spectral standard deviation around the one-sided centroid	finite scalar	physical_metric
crest_factor	ratio	audio peak divided by RMS with zero-RMS guard	finite scalar	physical_metric
mean_temporal_delta	normalized_pixel_difference	mean adjacent-frame absolute difference	finite scalar	physical_metric
sync_offset	ms	declared audio-minus-video offset	typed value	physical_metric

Objective metric results I

The metric definitions in tbl. 3 are computed from canonical or decoded artifacts and do not encode observer interpretations.

The default duck-rabbit artifact contains 3 unique raster levels and has mean normalized luminance 0.7770. These values are live generated facts, not perceptual effect sizes.

Observer estimands and verification controls I

Observer estimands and verification controls I

Table 4: Data-free observer outcomes and preregistered estimands.

Estimand	Response and model	Reference and contrast	Uncertainty
sifi flash count contrast	event count; poisson or ordinal	sifi one beep; two-beep minus one-beep event-count response	95% confidence interval
temporal binding offset contrast	continuous magnitude; linear mixed	temporal sync; reported timing shift per declared sync offset	95% confidence interval

Observer estimands and verification controls I

The analysis contract in tbl. 4 specifies estimands and uncertainty procedures without asserting any result.

Observer estimands and verification controls I

Table 5: Verification failure modes and negative controls.

Failure mode	Control	Expected result
altered encoded file	recompute encoded SHA-256	fail
stale canonical digest	recompute canonical little-endian float32 digest	fail
incorrect decoded dimensions	compare inspection facts with manifest summary	fail
stream timing mismatch	compare duration/timebase and declared sync offset	fail
unsupported backend	capability probe before encoding	explicit capability error
conflicting format requests	reject format_name/output_spec disagreement	parameter error

Observer estimands and verification controls I

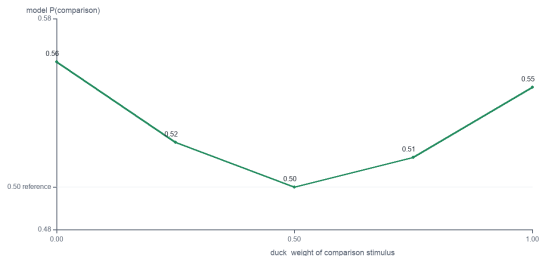
The negative controls in tbl. 5 test package integrity and media contracts; they are not null findings about perception.

Synthetic psychophysics model diagnostic I

Synthetic psychophysics model diagnostic I

Synthetic psychophysics: transparent feature observer

A deterministic model diagnostic varies duck_weight against a fixed reference; it is not a human or pretrained vision-model result.



Model contract

```
model = duckrabbt.synthetic.feature_observer
version = 1.0
temperature = 0.05
training_data = none
calibration = analytic_logistic
human_validation = not_performed

reference = duck_weight=0.50
points = 5
P range = 0.500~0.559
```

This curve diagnoses sensitivity of a specified feature observer. It does not estimate a psychometric function or establish a perceptual effect.

Figure 9: A narrow-range line plot shows deterministic model probability across duck_weight values beside the model identity, temperature, no-training-data statement, and human-data boundary.

Synthetic psychophysics model diagnostic I

What this figure shows: A narrow-range line plot shows deterministic model probability across `duck_weight` values beside the model identity, temperature, no-training-data statement, and human-data boundary. The hand-specified feature observer compares `duck_weight` variants with a fixed canonical reference. Its probability trace is computed analytically from serialized features, weights, and temperature; source data `output/data/synthetic_psychophysics.json` records model ID/version, reference and comparison digests, seed, calibration, `training_data=none`, and `human_data=false`. The y-axis is deliberately a narrow model-output range rather than a human psychometric scale. Controls: `duck_weight` comparison against fixed reference; serialized feature weights and temperature; seed `s=0`. Objective facts: model probability and score delta on the displayed model-output scale; canonical stimulus digests; `human_data=false`;

Synthetic psychophysics model diagnostic I

training_data=none. Claim level: synthetic_model_output.
Source data: output/data/synthetic_psychophysics.json (SHA-256 digest recorded in the figure registry). Evidence lineage: observer:preregistered_estimands. Limitations: The model is hand-specified, has no training data, has not been calibrated against observers, and does not estimate a human threshold or effect size. Its probabilities are model outputs, not participant responses or a psychometric function. Boundary: This diagnostic tests deterministic orchestration and metamorphic input-output behavior only; empirical psychophysics remains future work.

Synthetic psychophysics model diagnostic I

The diagnostic fig. 9 evaluates a fully serialized, hand-specified feature observer against a fixed canonical reference while varying `duck_weight`. Its logistic probabilities are model outputs generated from explicit features, weights, temperature, and seed; they are not a pretrained vision-model score, a human psychometric function, an assumed human effect size, or participant data. A future human study would require a preregistered task, calibrated display and playback, consented participants, exclusion and missingness rules, and an analysis contract before any observer claim could be made.